

Same Goal, Different Words: Spatial Instruction Sensitivity in World–Action Models

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Abstract: A robot should respond consistently to equivalent descriptions of a physical goal. We test three world–action models by naming either the moved object or the reference object first in a spatial relation. Across 864 matched rollouts on four RoboLab scenes, reference-first descriptions produce fewer stable spatial outcomes, with an average gap of 12.2 percentage points concentrated in particular relations. Reaching an arrangement also does not establish that the intended object was moved or that the arrangement will be retained. Evaluating world–action models therefore requires both alternative descriptions of the same goal and checks that distinguish reaching a placement from completing the instruction.

Keywords: World–Action Models, Spatial Language, Robot Evaluation

1 Introduction

World–action models (WAMs) generate robot actions together with predictions of future observations [1]. Their use for language-conditioned manipulation raises a basic question: how reliably do their actions reflect the requested goal? A robot must identify both which object to move and where to place it, and these choices should remain consistent across equivalent instructions.

Prior work evaluates both responsiveness to changed goals and consistency under equivalent instructions. TRI’s LBM study tests language conditioning in visually similar scenes [2]. CAST improves vision–language–action (VLA) instruction following by associating diverse descriptions with atomic commands and synthesizing actions for alternatives to recorded trajectories [3]. LIBERO-CF tests alternative goals in familiar scenes [4], while other evaluations examine language perturbations [5], meaning-preserving paraphrases from shared starts [6], and irrelevant context [7].

Within WAMs, SG-WAM studies instruction alignment and language perturbations [8]. We examine consistency under spatial reference reversal: “mustard right of box” and “box left of mustard” specify the same arrangement while both instructions require moving the mustard (Table 1). This tests agreement on a complete placement goal, allowing different action sequences to reach it.

We evaluate three released WAMs in RoboLab [9], holding the initial scene, requested placement, and object to move fixed while changing the relational description. We measure whether the arrangement is reached and retained, and examine cases where the spatial score disagrees with the intended manipulation. This comparison separates wording sensitivity from physical task difficulty and clarifies what placement success reveals about instruction following.

2 Paired spatial evaluation

Change the description, hold the goal fixed. We use four RoboLab scenes: cube and bowl, butter and raisin box, mustard and raisin box, and two bowls. Goals are left/right/front/behind for the cube, left/right/supported placement on top for butter and mustard, and both bowl-stacking orders. Seven of the twelve goals come from RoboLab; we add five placements using the same scenes and objects.

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Table 1: One goal, three instructions. Separate runs use the same model, starting scene, requested placement, and model settings.



Example goal: mustard to the right of the box.

Target: the mustard bottle, which the robot should move.

Reference: the raisin box, which defines the destination.

Instruction form	Exact tested instruction
Direct (D)	Place the mustard bottle on the table to the right of the raisin box.
Mover-first (S)	Move the mustard bottle on the table so that the mustard bottle is to the right of the raisin box.
Reference-first (I)	Move the mustard bottle on the table so that the raisin box is to the left of the mustard bottle.
S versus I	Does describing the same placement through the reference object change the outcome?
D versus S	Does expanding the direct instruction change the outcome?

Table 1 defines the three instruction forms. Direct (D) uses RoboLab wording where available. Mover-first (S) and reference-first (I) refer to the object named first after “so that,” with the relation reversed in I. All forms explicitly ask to move the same object.

For supported placement, “on top of and supported by” becomes “underneath and supporting.” For bowls, the latter does not explicitly require the nesting used by the success check. We show those results separately.

Compare models from the same starts. We evaluate the DROID versions of Cosmos3 Nano [10], Cosmos3 Edge [11], and FLUX 3 Action [12]. Nano and FLUX perform strongly on RoboLab’s leaderboard; Edge provides a smaller Cosmos-family comparison [13]. Inputs are one wrist and two exterior cameras plus robot state. After small object-position and orientation changes, we retain the first eight starts per scene solvable by a scripted controller. Every model runs every goal and description from these starts: 864 episodes, with identical initial states and observations across paired runs.

Policies execute 32 actions at a time at 15 Hz, using the same inference seed and fixed settings for each model. Each episode lasts 20–40 s, depending on the scene, and continues after the first successful placement. Full instructions and evaluation settings accompany the paper.

Separate reaching a placement from completing the task. We count a *stable spatial outcome* when, for at least one second, the requested arrangement holds, the target object has the required support and moves slower than 2 cm/s, and the gripper is no longer touching it. Spatial directions are defined relative to the robot. This outcome can occur at any point in the episode. An additional analysis checks whether it holds during the final second. Neither score verifies which object was moved: moving the reference may also make the relation true.

For each matched pair, we subtract the reference-first outcome (1 if reached, 0 otherwise) from the mover-first outcome: $S - I$. Positive differences favor mover-first wording. $D - S$ tests expanding the direct instruction. We average over goals within each scene, then equally across scenes and models, reporting gaps in percentage points.

Cube-right, butter-right, and mustard-left hold initially and pass under all instructions, contributing zero to the at-any-time differences. Goals requiring a new arrangement are shown separately. Confidence intervals resample starts within scenes, keeping models, goals, and descriptions paired. Sign-flip tests use Holm correction for the two comparisons.

3 Results

Reference-first descriptions produce fewer stable outcomes. Naming the target first in the spatial relation leads to more stable outcomes than naming the reference first, with an average gap of 12.2 percentage points (95% interval: 8.6–15.4; adjusted $p < 0.001$). The average has the same sign for all three models (Figure 1), while the direct-versus-expanded contrast is smaller and inconclusive. The additional check at the end of each episode gives a similar reference-inversion gap, so the finding is not explained solely by transient placements.

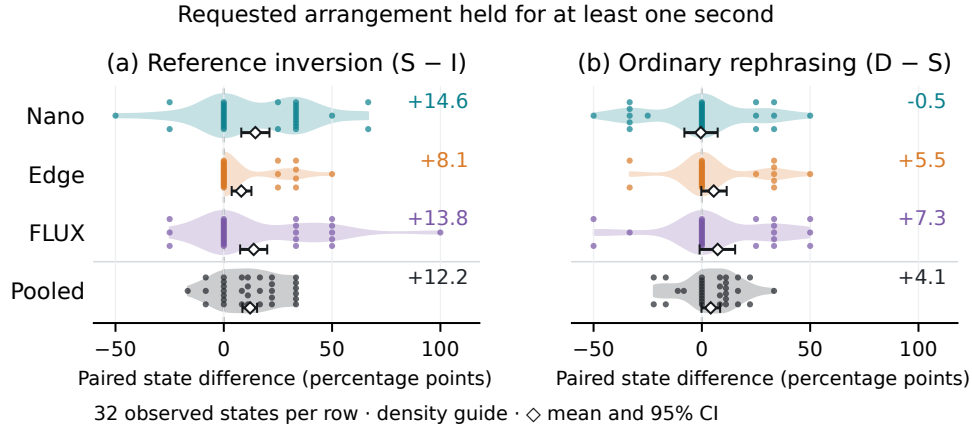


Figure 1: Paired wording effects. Positive S–I favors mover-first descriptions; D–S compares direct and expanded wording. Each dot averages paired differences over goals within one start (also over models in the pooled row). Shading guides the eye; diamonds and bars show means and 95% state-bootstrap intervals. Model-level intervals are descriptive.

A few relations drive the average. Mustard-right produces the largest pooled gap, with the same direction in all three models. The pooled descriptive gap remains positive when bowl stacking is excluded. The cube scene, however, has no average S/I difference, and cube-behind favors reference-first wording under the recorded metric (Figure 2).

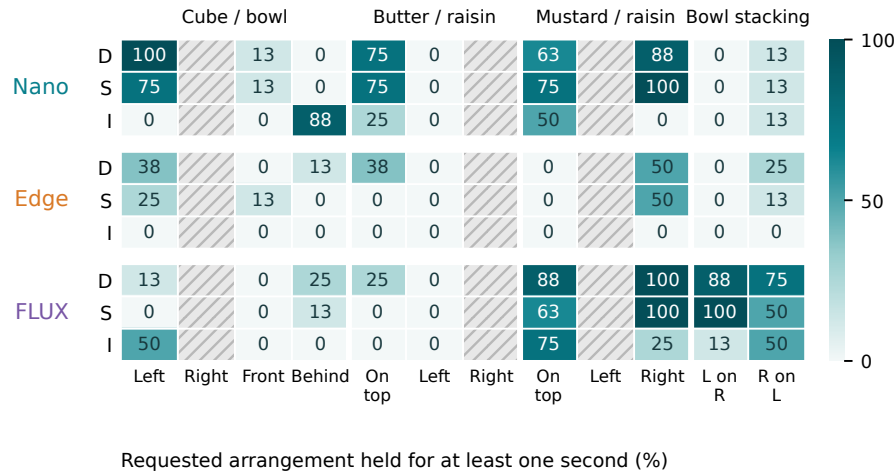


Figure 2: Stable outcomes by goal and wording, with eight starts per cell. Hatching marks initially satisfied goals. Scores do not verify complete instruction following.

A Edge · mustard right of the raisin box



B Nano · cube behind the bowl

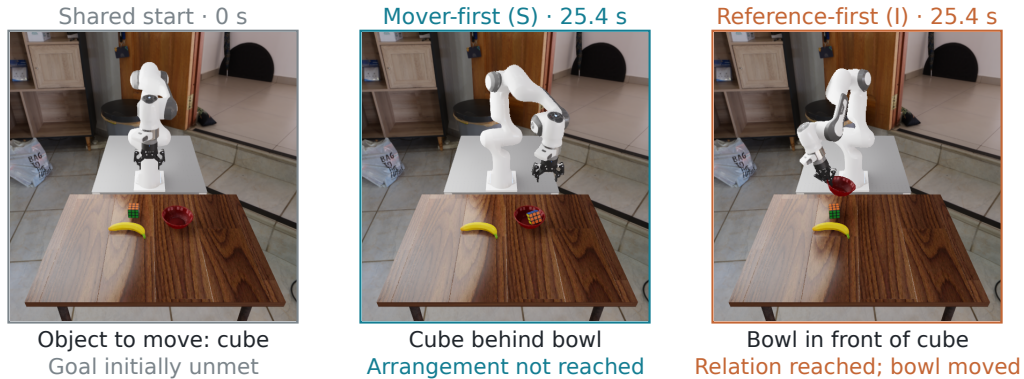


Figure 3: Matched executions under two descriptions of the same goal. (a) Edge reaches mustard-right only with mover-first wording. (b) Nano reaches the scored cube-behind relation while moving the bowl; the relation is lost by episode end. Snapshots use the same times within each pair. A shared crop and rotation improve visibility; directions refer to the robot. Examples were selected for their contrasting scores.

4 Discussion

An instruction specifies an arrangement and which object should move. Equivalent descriptions should preserve both requirements. Following controlled simulator evaluation [14], shared starts rule out scene difficulty as the sole explanation for our wording gap, but cannot distinguish wrong-goal selection, wrong-object selection, and execution failure. LIBERO-CF separately scores target-object contact and task success [4]; our placement score needs the same care in interpretation.

Figure 3 illustrates why a higher placement score need not mean better instruction following: Nano reaches the relation by moving the bowl while the cube remains on the table. Roughly half of all episodes with a stable outcome also lose it before the end. Evaluation should distinguish reaching the arrangement, performing the intended manipulation, and retaining it under continued control.

CAST improves instruction following through counterfactual training in visual navigation [3], while SG-WAM supplies semantic guidance for manipulation [8]. Whether these approaches preserve behavior under equivalent spatial descriptions deserves testing. For WAMs, does wording change the predicted goal, or does execution depart from an appropriate prediction? Answering requires aligned predictions and executions, plus separate changes to relation vocabulary and argument order. Our evidence establishes neither a WAM-specific mechanism nor a disadvantage relative to VLAs. Bowl support/containment ambiguity further limits claims of equivalent instructions.

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